HAMS DATA SCIENCE CHALLENGE · BAYESIAN MARKETING MIX MODEL

Which ad channels really drive sales?

Nishant Singh

The data covers **104 weeks** of revenue and spend across seven paid channels for an online shop. In total, about **7.2 million** was spent and about **14.2 million** came in as revenue. I built a Bayesian model in PyMC that links weekly revenue to the carried-over spend of each channel. **Channel 5** has the best return at **1.34**.

QUESTION 1 · SPEND CARRY-OVER

An ad keeps working after the week it runs, so I used adstock

Adstock captures this carry-over. This week's spend counts in full. Last week's spend counts for a fraction of its value, called α. The week before counts for α times α, and so on for **eight weeks**. Each channel has its own α. For example, if α is 0.3, last week's ad still counts for 30% and the week before for 9%. Weekly revenue is then modelled as a baseline of normal sales, a trend, a yearly pattern and Black Friday, plus the carried-over effect of each channel. My α values came out between **0.26** and **0.33**. The data barely moved them from my starting guess, so the carry-over speed is mostly an assumption on my part.

QUESTION 2 · CHOICE OF PRIORS

I chose wide but sensible starting guesses and tested them before fitting

A prior is a starting guess about a number before the model sees any data. I kept my guesses wide enough for the data to decide, but set sensible limits. For example, an ad cannot lower sales. To test the guesses, I simulated revenue from them alone. My first set gave a middle value of **1.64** against a real value of **0.33**, which is about five times too high. I lowered the guesses for channel effects, baseline and seasonality. The final simulations ranged from **0.04** to **0.87**, while real revenue ranges from **0.15** to **1.0**.

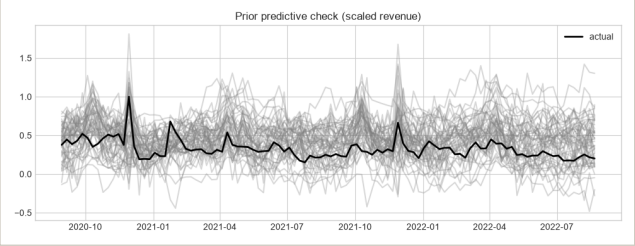


Figure 1. Revenue simulated from my final guesses in grey, against real revenue in black.

PARAMETER	STARTING GUESS	WHY
α, carry-over	Beta, 2 and 4	Stays between 0 and 1 and allows short and long effects.
β, channel effect	Half normal, scale 0.1	Ads cannot reduce sales, and seven channels together should not exceed total revenue.
Baseline level	Normal, centre 0.2, spread 0.1	Normal sales without ads, as a share of the maximum.
Trend and seasonality	Normal, centre 0, spread 0.2 and 0.1	Starts with no trend and prevents large artificial yearly waves.
Black Friday	Normal, centre 0, spread 0.3	Lets those weeks rise well above normal.
Noise	Half normal, scale 0.1	Small, positive variation from week to week.

QUESTION 3 · PRIOR AND POSTERIOR SAMPLING

Prior sampling tests my guesses and posterior sampling shows what the data taught the model

Prior sampling simulates results from my guesses alone, with no data. I used it to check that the guesses made sense, as shown in Figure 1. **Posterior sampling** updates those guesses using the real 104 weeks of data. I used the NUTS sampler with four runs of 1,000 draws each. All results on revenue, effects and ROI come from the posterior. The four runs agreed closely. R-hat was **1.003**, where anything below 1.01 is good. The smallest effective sample size was **2,154**, where anything above 400 is good. There were **no divergence warnings**.

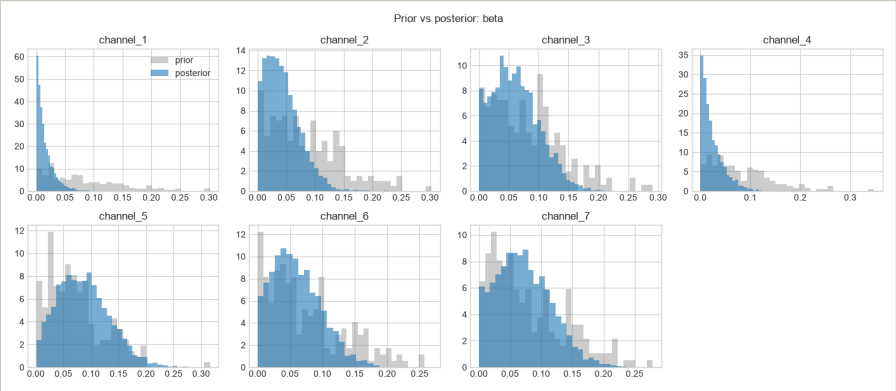


Figure 2. Effect β per channel, starting guess in grey and result after the data in blue.

Figure 2 shows each channel's effect β, in grey before the data and in blue after. For channels 1, 2 and 4, the data pulled β close to zero. Channel 5 moved furthest from zero, to **0.087**. β is the revenue a channel adds per unit of spend on the model's 0 to 1 scale. It is not the same as ROI, because the channels spend very different amounts.

QUESTION 4 · MODEL QUALITY

The model predicts new weeks reasonably well but misses one-off spikes

I used three measures. **R²** shows how much of the weekly ups and downs the model explains, where 1 is perfect. **MAPE** is the average percentage error. **Coverage** shows how often real revenue falls inside the model's 94% range, and it should be close to 94%. I also trained the model on the first **88 weeks** and asked it to predict the last **16**, which it had never seen.

MEASURE	RESULT	WHAT IT MEANS
R², all 104 weeks	0.63	Explains 63% of the weekly changes in revenue.
MAPE, all 104 weeks	17.7%	A typical miss of about 24k on an average 136k week.
Coverage of the 94% range	97.1%	101 of 104 weeks fall inside the range, so the model is slightly cautious.
MAPE, 16 unseen weeks	11.7%	About 12% average error on weeks the model had never seen.
R², 16 unseen weeks	0.06	Low because those weeks are flat, so MAPE is the fairer measure.

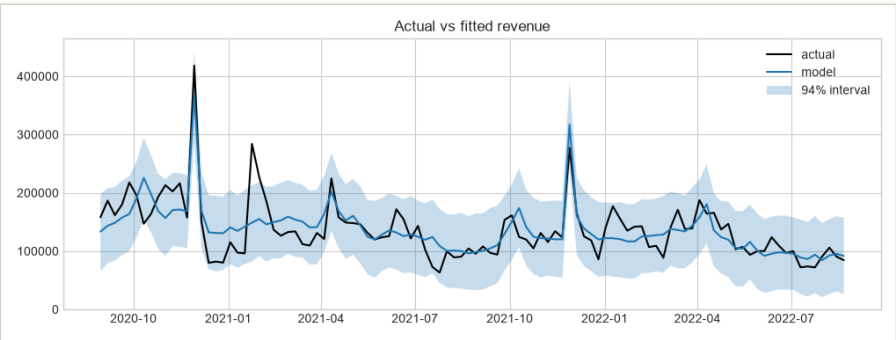


Figure 3. Real revenue in black, the model in blue and its 94% range shaded.

The model follows the downward trend and both Black Friday peaks. It misses a few spikes that have no matching change in spend. In late January 2021, revenue was about **285k** but the model predicted about **150k**. This may have been a promotion, but I cannot confirm that. I have not compared the model with a simple benchmark, so I cannot say how much better it is than a basic guess.

QUESTION 5 · CHANNEL PERFORMANCE AND EFFECTS

Channels 7 and 3 take 68% of the budget but bring in 22% of revenue

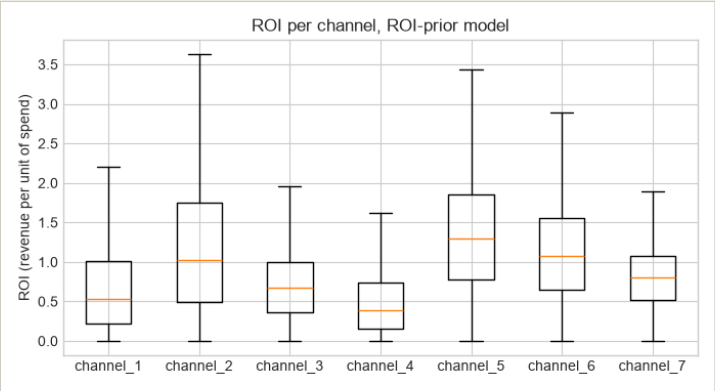
Paid ads bring in about **44%** of revenue. Normal sales, the trend and Black Friday bring in the other **56%**. Channel 5 has the strongest effect, with a β of 0.087, while channels 1 and 4 have the weakest. Every channel's effect is positive, but every range starts close to zero, so no effect is pinned down sharply. Channel 7 alone takes about **40%** of the spend. Together, all channels return about **0.85** for every 1 spent.

QUESTION 6 · ROI AND THE BEST CHANNEL

Channel 5 ranks first, but only by a small margin

ROI is the revenue a channel brought in divided by what was spent on it. An ROI of 1.0 means break even on revenue, before product costs. I worked it out for every posterior sample, so each channel gets a range instead of a single number. My first model made the tiny channel 2 look amazing, with an ROI of **17.68**. This was a trap. Giving every channel the same guess for β quietly implies a huge ROI for a channel with very little spend. So I repeated the model with the guess placed on ROI itself. Channel 2 fell to **1.22**. The table shows this second model.

CHANNEL	SPEND SHARE	REVENUE SHARE	ROI	94% RANGE	CHANCE BEST	FIRST MODEL
5	**12.4%**	**9.3%**	**1.34**	**0.11 to 2.88**	**32.9%**	**1.47**
2	0.5%	4.5%	1.22	0.07 to 3.29	25.6%	17.68
6	7.3%	4.0%	1.14	0.12 to 2.42	21.8%	1.08
7	39.9%	12.0%	0.82	0.12 to 1.63	5.8%	0.59
1	1.8%	1.0%	0.71	0.03 to 2.22	8.0%	1.11
3	28.1%	10.2%	0.71	0.04 to 1.63	3.4%	0.72
4	10.0%	3.1%	0.52	0.02 to 1.66	2.6%	0.62



Channel 5 is the best guess at 1.34 and is first or near first in both models, but it has only a **32.9%** chance of really being first. Channels 2 and 6 are close behind, and channels 3, 1 and 4 look weakest. If the budget has to move, the evidence mildly supports small tests that shift money from channels 3, 4 and 7, which take about 78% of spend, towards channels 5 and 6.

HOW FAR TO TRUST IT AND WHAT NEXT

The model assumes every extra unit of spend returns the same amount. In reality, returns shrink as spend grows, so the ROI here is an average and not what the next euro would earn. It also counts revenue, not profit. With only 104 weeks and channels that partly move together, the model cannot cleanly separate every effect. It shows association, not proof of cause.

1. Run small on and off ad tests to measure real incrementality.
2. Let returns shrink as spend grows and let effects change over time.
3. Find promotion data to explain the unexplained spikes.
4. Compare the model against a simple benchmark.